

Docker Exam

Time: 2 hours

Full Marks: 50

Group A

The answers should be in a word file or as a pdf and write the answers with the commands and screen-shots attached like you build assignments.

2 marks questions: (Any 10)

1. VM vs docker.
2. Pros and cons of docker.
3. What is volume in docker and what is mounting.
4. Image vs containers.
5. What is docker network.
6. Create a network and run a busybox os container inside it and tell the network address and the container address(Need just the commands).
7. Create two busybox instances e.g. b1 and b2 and ping each other buy name(Attach the text outputs and add commands).
8. Run a mysql db container with port mapping and 1-2 lines on what is port mapping/port forwarding/port binding(Attach the commands too).
9. Why Dockerfile is needed?
10. Why docker-compose.yaml file is needed?
11. Why .dockerignore file is needed?
12. What is hypervisor?

Group B

5 marks questions: (Any 6)

1. Write an express server code create an image out of it run a container from it(port mapping) curl from the host machine and write the success output.
2. Create a volume v1 attach to container c1 and persist some data, now remove the container c1 and create another container c2 and attach the volume v1 to it and show that the data is still there.
3. Run a mysql database container and jump inside into it and create a database then a table and insert some data into the table and apply select * from <table-name>.(Write the commands and attach some screen shots).
4. Run a multi container application nodejs server + mysql database using compose.yaml file.
5. Create an image(your application code) publish to docker hub(I will pull it and run a container).
6. Explain docker networks with full details and examples.
7. Run an express server container that will send an index.html file in response of a request at '/' route(Good index.html file will be appreciated).
8. create any * pattern code in c and create an image of it run a container that will print the * pattern.